

 RATIONALE		PLG-RTNL-0005-V1	
TPL-001-4 Contingencies Rationale		Revision:	1
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1. Overview

1.1 Purpose

The following rationale is to document consistent contingency description and selection to assure ATC compliance with the NERC TPL-001-4 Reliability Standard for the Bulk Electric System (BES).

1.2 Scope

This document provides the rationale for meeting the NERC Reliability Standard TPL-001-4 Requirements 2.6, 3.4, 3.5, 4.4, and R4.5.

2. Applicability

2.1 Regulatory Applicability

NERC Reliability Standard TPL-001-4 requires ATC compliance as a registered Transmission Planner.

2.2 Affected Assets

ATC BES assets.

2.3 Affected Personnel

Generally System Planning department transmission planners and management.

3. Specific Rationale/Considerations

The specific rationale/considerations that are used to develop the lists of contingencies that are used for ATC's reliability analyses are given below. Different contingency lists are created for each planning horizon power flow model that is used for ATC's reliability analyses.

The specific rationale and considerations that ATC will use to assure compliance with the NERC TPL-001-4 standard are described below.

The TPL-001-4 requirements that are particularly pertinent to this ATC rationale are the following:

R2.6 - For steady state, short circuit, or Stability analysis, past studies may be used to support the Planning Assessment if:

R2.6.1 – the study is five calendar years old or less, unless a technical rationale can be provided to demonstrate that the results of an older study are still valid

R2.6.2 –no material changes have occurred to the System represented in the study.

Documentation to support the technical rationale for determining material changes shall be included.

The supporting analyses for TPL-001-4 assessments are performed primarily by ATC. As explained in RSAW documentation in support of R2.6, ATC performs various studies and simulations each year to support its 10-Year Assessment. ATC typically relies heavily on the current and past two years of studies and system simulations for the current 10-Year Assessment.

However, the analysis results of other functional entities, such as MISO, MRO, RF, and neighboring Transmission Planners may also be used to support compliance with TPL-001-4 requirements and the ATC Planning Criteria. When the results of other entities are used, ATC will maintain appropriate documentation to support the use of these results.

R3.4 – Those planning events in Table 1, that are expected to produce more severe steady state System impacts on the ATC portion of the BES shall be identified and a list of those Contingencies to be evaluated for System performance in Requirement R3, Part 3.1 created. The rationale for those contingencies selected for evaluation shall be available as supporting information.

R3.5 – Those extreme events in Table 1 that are expected to produce more severe System impacts shall be identified and a list created of those events to be evaluated in Requirement R3, Part 3.2. The rationale for those Contingencies selected for evaluation shall be available as supporting information. If the analysis concludes there is Cascading caused by the occurrence of extreme events, an evaluation of possible actions designed to reduce the likelihood or mitigate the consequences and adverse impacts of the event(s) shall be conducted.

R4.4 – Those planning events in Table 1, that are expected to produce more severe System Stability impacts on the ATC portion of the BES shall be identified and a list of those Contingencies to be evaluated for System performance in Requirement R4, Part 4.1 created. The rationale for those contingencies selected for evaluation shall be available as supporting information.

R4.5 – Those extreme events in Table 1 that are expected to produce more severe System impacts shall be identified and a list created of those events to be evaluated in Requirement R4, Part 4.2. The rationale for those Contingencies selected for evaluation shall be available as supporting information. If the analysis concludes there is Cascading caused by the occurrence of extreme events, an evaluation of possible actions designed to reduce the likelihood or mitigate the consequences of the event(s) shall be conducted.

ATC performs and evaluates the planning and extreme events in Table 1 that would produce the more severe System impacts. The ATC rationale below provides the basis and explanation of the contingencies that are selected as being the more severe and subject to simulation (e.g. TPL-001-4 Contingency Lists).

TPL-001-4 Table 1 Events

This document provides ATC's rationale for each type of contingency event listed in Table 1. In general, this rationale is formatted with a discussion of the basis for the steady state contingency selections and then the basis for dynamic (stability) contingency selections for each planning event Category in Table 1. The extreme events rationale is constructed similarly and follows the planning events discussion.

ATC maintains a Master Contingency List based on ATC's regularly updated As-Built transmission system model. Planned changes to the As-Built system are entered into the master list as future

alterations or additions to the master list. The planned changes can be enabled or disabled based on a status flag, if for some reason the model that a contingency file is to be created for has intentionally excluded that project. ATC's annual 10-Year Assessment (TYA) study models are generally the basis for the models used in ATC's annual Planning Assessments. For each TYA, a copy of the Master Contingency List file is saved for the Project Deficient models and a separate Master Contingency List file is saved for the All-Projects models based on the projects selected to be included in the models. As new projects are completed, contingencies created (representing the project's impact on the system) are reviewed for correctness and then they are committed to the Master Contingency List file. A check for errors in the contingency list is performed prior to finalizing the contingency lists to be used for the specific TYA study models.

ATC has developed a tool that it uses to extract a comprehensive list of any specific contingency type from this master list. The in-service date for each future project to be represented in a particular System model is taken into account by the tool when generating study-specific TPL-001-4 contingency lists. For the steady state planning event analyses required by TPL-001-4, these comprehensive contingencies lists may be further refined according to the rationale described in the rest of this document.

ATC does not have any BES Elements with a voltage rating above 345 kV. So, the EHV performance requirements in Table 1 only apply to 345 kV elements.

Stability analysis is "transient dynamic" analysis (which will simply be referred to as "dynamic" analysis in this document) of angular stability or voltage stability.

- Dynamic angular stability is the circumstance where the angle difference between one or more BES generating units and the transmission system is small enough that the units are expected to remain synchronized to the system and the angular difference oscillations exhibit acceptable damping following a contingency event.
- Dynamic voltage stability is the circumstance where the voltage of one or more BES buses is not expected to result in the tripping of any system load or BES generating units.

For dynamic angular stability analysis of any Table 1 event, the selected contingencies are ones that are expected to cause more severe transient angular disturbances to one or more generator units. Generally, more severe impacts are expected for contingencies that affect generating units at a generating plant with total net output of 100 MW or more and that are interconnected to the BES. In addition, the selected contingencies are included in generator-centric contingency lists in any given year if one or more of the following "normal angular stability inclusion" criteria are also met:

- A significant system topological change occurred near the generator unit
- A significant generator model type or parameter value change occurred.
- The last angular stability analysis with the contingency is more than 5 years old.

For dynamic angular analysis, the more severe contingencies are those in the vicinity of the generator under study. The contingencies elsewhere in the ATC system have a less severe impact on dynamic stability because either the fault location is further away electrically (i.e. less accelerating power) or the post-contingency power-angle curve (i.e. maximum power transfer capability curve) is larger (i.e. lower system Thevenin equivalent impedance as seen from the generator) than for those contingencies evaluated by ATC. So, the contingency lists for dynamic angular stability analysis are more selective and generator-centric, as opposed to the steady state contingency list, which is comprehensive and system-centric.

For dynamic voltage stability analysis of any Table 1 event, the selected contingencies are ones that are expected to cause more severe transient minimum or maximum voltages at BES buses in an weak system area that has been identified using other studies.

For dynamic voltage analysis, the more severe contingencies are those in the vicinity of a weak system area under study. The contingencies elsewhere in the ATC system have a less severe impact on dynamic voltage stability because the fault location is further away electrically. So, the contingency lists for dynamic voltage stability analysis are more selective and study area-centric, as opposed to the steady state contingency list, which is comprehensive and system-centric.

The contingency types selected for dynamic voltage analysis area-centric contingency lists may be overridden or reduced when steady state analysis indicates that certain types of contingencies would be more severe.

Section 3.1 Planning Event Contingencies

P1 (Single Contingency – loss of one Transmission Element)

These contingencies represent a three phase fault on a single selected BES Element with normal clearing. These outages are contingencies with the initial condition of a normal system. All of the P1 contingencies are 'no load loss allowed' contingencies. These outages are all event-based contingencies. The definition of event-based contingencies is given in Section 4.1 below.

P1.1 (Single contingency – loss of a generator)

These contingencies represent a three phase fault on the terminal of a BES generating unit with normal (primary) clearing. The types and combinations of generating units that are treated as a single "generator" are defined in the Section 4.1 below.

Steady state analyses

Every year ATC generates a comprehensive list of applicable P1.1 contingencies, which includes every BES generating unit that is interconnected to the ATC system.

The P1.1 contingencies outside of the ATC system are those that are part of MISO's annual P1.1 MTEP reliability assessment that include monitoring the ATC system.

Dynamic analyses

Angular stability analysis

No P1.1 angular stability contingencies are included in the generator-centric lists. Angular instability is more severe for comparable P1.2 or P1.3 contingencies because the application of a fault that is electrically closer to the study generator and may remove ties to the surrounding BES network without outaging any other generators is more severe than a fault on the terminal of a nearby BES generator. Therefore, each applicable P1.1 contingency is covered by a comparable P1.2 or P1.3 contingency.

Other online BES generation in the ATC system is more stable when a BES generating unit is outaged any other fault (e.g. P1.2, P1.3, etc.).

Voltage stability analysis

No P1.1 voltage stability contingencies are included in the study area-centric lists. Voltage instability is more severe for the ATC system area under study for comparable P1.3 contingencies because the application of a fault on the high side of a BES generator's GSU transformer outages the generator, is electrically closer to the study area, may remove strong ties to the surrounding BES network or may

island the study area. Therefore, each applicable P1.1 contingency is covered by a comparable P1.3 contingency.

The P1.1 dynamic contingencies outside of the ATC system are those that are part of MISO's annual P1.1 MTEP reliability assessment that include monitoring the ATC system.

P1.2 (Single contingency – loss of one transmission circuit)

These contingencies represent a three phase fault on a BES line with normal (primary) clearing.

Steady state analyses

Every year ATC generates a comprehensive list of applicable P1.2 contingencies, which includes every BES line in the ATC system. The Mackinac back-to-back HVDC facility contingencies are included in this list. The HVDC facility is a series element of a 138 kV network line. The outage of this 138 kV network line is represented by three different contingencies – outage of the entire facility (both converters), outage of the south converter (north converter in shunt device mode), and outage of the north converter (south converter in shunt device mode). Each of these three contingencies simulate the loss of network line flow (see Appendix).

The P1.2 contingencies outside of the ATC system are those that are part of MISO's annual P1.2 MTEP reliability assessment that include monitoring the ATC system.

Dynamic analyses

Angular stability analysis

The P1.2 contingencies that are included in the generator-centric list are those that will cause more severe angular instability for a BES generator connected to the ATC system that is under study than any other type of contingency. Angular instability is more severe for P1.2 faults that are applied on the end of BES lines in the ATC system that are electrically closer to the study generator and may remove ties to the surrounding BES network without outaging any other generators.

Voltage stability analysis

The P1.2 voltage stability contingencies in the ATC system that are included in study area-centric lists are those for which steady state screening studies indicate the potential for voltage instability.

The P1.2 dynamic contingencies outside of the ATC system are those that are part of MISO's annual P1.2 MTEP reliability assessment that include monitoring the ATC system.

P1.3 (Single contingency – loss of one transformer)

These contingencies represent a three phase fault on a BES transformer with normal (primary) clearing.

Steady state analyses

Every year ATC generates a comprehensive list of applicable P1.3 contingencies, which includes every BES transformer circuit in the ATC system.

The P1.3 contingencies outside of the ATC system are those that are part of MISO's annual P1.3 MTEP reliability assessment that include monitoring the ATC system.

Dynamic analyses

Angular stability analysis

The P1.3 contingencies that are included in the generator-centric list are those that will cause more severe angular instability for a BES generator connected to the ATC system that is under study than any other type of contingency. Angular instability is more severe for P1.3 faults that are applied on the end of BES transformer circuits that are electrically closer to the BES generator connected to the ATC system that is under study and may remove ties to the surrounding BES network without outaging any other generators. GSU transformers are only included on the list to cover comparable P1.1 contingencies.

Voltage stability analysis

The P1.3 voltage stability contingencies in the ATC system that are included in study area-centric lists are those for which steady state screening studies indicate the potential for voltage instability.

The P1.3 dynamic contingencies outside of the ATC system are those that are part of MISO's annual P1.3 MTEP reliability assessment that include monitoring the ATC system.

P1.4 (Single contingency – loss of a shunt device)

These contingencies represent a three phase fault on a BES shunt device with normal (primary) clearing. Multiple shunt device outages are modeled as a single event if they are connected to the bus through the same breaker. Shunt devices include capacitor banks, inductor banks, and shunt power electronic devices.

Steady state analyses

Every year ATC generates a comprehensive list of applicable P1.4 contingencies, which includes every BES shunt device in its system. Power electronic shunt devices connected to the ATC system, such as the Benson Lake SVC, are included in the contingency list. The north and south converters of the Mackinac back-to-back HVDC facility act as two independent shunt devices (see Appendix). The contingencies of the loss of either one or both converters (shunt devices) are already included in the P1.2 contingencies and do not have to be duplicated.

The P1.4 contingencies outside of the ATC system are those that are part of MISO's annual P1.4 MTEP reliability assessment that include monitoring the ATC system.

Dynamic analyses

Angular stability analysis

No P1.4 angular stability contingencies in the ATC system are included in the generator-centric list. P1.4 shunt device contingencies will not cause more severe angular instability of generators than any of the other P1.2 or P1.3 contingencies because shunt device contingencies do not remove any ties to the surrounding BES network. Therefore, each applicable P1.4 contingency is covered by a corresponding P1.2 or P1.3 contingency.

Voltage stability analysis

The P1.4 voltage stability contingencies in the ATC system that are included in study area-centric lists are those for which steady state screening studies indicate the potential for voltage instability.

The P1.4 dynamic contingencies outside of the ATC system are those that are part of MISO's annual P1.4 MTEP reliability assessment that include monitoring the ATC system.

P1.5 (Single contingency – loss of a pole of a DC line)

These contingencies represent a single-line-to-ground fault on a pole of a BES DC line with normal (primary) clearing. The Mackinac back-to-back HVDC facility does not apply to this category because ATC interprets that there is no DC transmission line within the HVDC building (see Appendix).

Steady state analyses

There are presently no DC lines in the ATC system. Therefore, there are no P1.5 contingencies within the ATC system.

The P1.5 contingencies outside of the ATC system are those that are part of MISO's annual P1.5 MTEP reliability assessment that include monitoring the ATC system.

Dynamic analyses

Angular and Voltage stability analysis

There are presently no DC lines in the ATC system. Therefore, there are no P1.5 dynamic contingencies in the generator-centric or study area-centric lists.

The P1.5 dynamic contingencies outside of the ATC system are those that are part of MISO's annual P1.5 MTEP reliability assessment that include monitoring the ATC system.

P2 (Single Contingency – selected Transmission Element outage events)

These contingencies represent opening a line section without a fault or a single-line-to-ground (SLG) fault on a single select transmission Element with normal clearing. These outages are contingencies with the initial condition of a normal system.

P2.1 (Single Contingency - Opening of a line section without a fault)

These contingencies represent opening of a line section in a normally networked BES line without a fault. This category evaluates all normally networked lines that have one or more taps. Each open line section is modeled as the opening of only one end of the line with a protective switching device. All P2.1 contingencies are 'no load loss allowed' contingencies.

Steady state analyses

No load loss allowed - every year ATC generates a comprehensive list of applicable P2.1 contingencies that involve the opening of applicable BES lines sections. For any given transmission line, the more severe line section contingencies are selected. These more severe contingencies are the opening of one of the breakers at each of the ends of a normally networked transmission line (two separate P2.1 contingencies for two terminal lines, three separate P2.1 contingencies for three terminal lines). In lines with more than two sections between breakers these selected contingencies are more severe because all of the load from the taps on the line has to be served from the remote ends, creating the largest flows through the ends of the line that remain closed.

The P2.1 contingencies outside of the ATC system are those that are part of MISO's annual P2.1 MTEP reliability assessment that include monitoring the ATC system.

Dynamic analyses

Angular and Voltage stability analysis

The P2.1 contingencies do not explicitly cause either angular or voltage instability. Therefore, there are no P2.1 dynamic contingencies in the generator-centric or study area-centric lists.

P2.2 (Single Contingency - Bus section fault)

These contingencies represent a single-line-to-ground fault on a BES bus section with normal (primary) clearing. These outages are all event-based contingencies.

Steady state analyses

No load loss allowed - every year ATC generates a comprehensive list of applicable P2.2 contingencies, which includes every EHV (345 kV) bus section not connected to a transformer within the ATC system. ATC transformer secondary or tertiary voltages are all less than 300 kV.

Load loss allowed - every year ATC generates a comprehensive list of all applicable P2.2 contingencies from the current version of the master contingency list. The P2.2 contingencies outside of the ATC system are those that are part of MISO's annual P2.2 MTEP reliability assessment that include monitoring the ATC system.

Dynamic analyses

Angular stability analysis

No load loss allowed – No P2.2 EHV angular stability contingencies in the ATC system are included in the generator-centric lists. Angular instability is more severe for the BES generator under study for corresponding P4.5 or P5.5 contingencies because these contingencies outage an additional bus section and the delayed clearing causes a more severe system transient. Therefore, each applicable P2.2 contingency is covered by a corresponding P4.5 or P5.5 contingency.

Load loss allowed - The P2.2 HV angular stability contingencies are treated the same as the EHV contingencies.

Voltage stability analysis

No load loss allowed –The P2.2 voltage stability contingencies in the ATC system that are included in study area-centric lists are those for which steady state screening studies indicate the potential for voltage instability.

Load loss allowed - The P2.2 HV voltage stability contingencies are treated the same as the EHV contingencies.

The P2.2 dynamic contingencies outside of the ATC system are those that are part of MISO's annual P2.2 MTEP reliability assessment that include monitoring the ATC system.

P2.3 (Single Contingency – non-Bus-tie Breaker internal fault)

These contingencies represents a single-line-to-ground internal fault on a BES non-bus-tie breaker with normal (primary) clearing. These outages are all event-based contingencies. The identification of a non-Bus-tie Breaker can be determined from the definition of Bus-tie Breaker given in Section 4.1.

Steady state analyses

No load loss allowed - every year ATC generates a comprehensive list of applicable P2.3 contingencies. Based on footnote 1, ATC only includes every EHV (345 kV) non-Bus-tie breaker not connected to a transformer within the ATC system because ATC does not have any transformer low - side ratings exceeding a nominal 300 kV.

Load loss allowed - every year ATC generates a comprehensive list of all applicable P2.3 contingencies from the current version of the master contingency list. When two or more individual non-bus-tie breaker faults at a station result in the outage of the same elements, then only one contingency is included in the list to avoid redundant analysis.

The P2.3 contingencies outside of the ATC system are those that are part of MISO's annual P2.3 MTEP reliability assessment that include monitoring the ATC system.

Dynamic analyses

Angular stability analysis

No load loss allowed – No P2.3 EHV angular stability contingencies in the ATC system are included in the generator-centric lists. Angular instability is more severe for the BES generator under study for corresponding P4.1, P4.2, P4.3, P4.4, and P4.5 contingencies because they include delayed clearing due to a stuck non-Bus-tie-Breaker. Therefore, each applicable P2.3 contingency is covered by a corresponding P4.2, P4.3, P4.4, or P4.5 contingency.

Load loss allowed - The P2.3 HV contingencies are treated the same as the EHV contingencies.

Voltage stability analysis

No load loss allowed – The P2.3 EHV voltage stability contingencies in the ATC system that are included in study area-centric lists are those for which steady state screening studies indicate the potential for voltage instability.

Load loss allowed – The P2.3 HV voltage stability contingencies in the ATC system that are included in study area-centric lists are those for which steady state screening studies indicate the potential for voltage instability.

The P2.3 dynamic contingencies outside of the ATC system are those that are part of MISO's annual P2.3 MTEP reliability assessment that include monitoring the ATC system.

P2.4 (Single Contingency – Bus-tie Breaker internal fault)

These contingencies represent a single-line-to-ground internal fault on a BES Bus-tie breaker with normal (primary) clearing. These outages are all event-based contingencies. All of P2.4 contingencies are 'load loss allowed' contingencies. The identification of a Bus-tie Breaker can be determined from the definition of Bus-tie Breaker given in Section 4.1.

Steady state analyses

Load loss allowed - every year ATC generates a comprehensive list of applicable P2.4 contingencies from the current version of the master contingency list. The P2.4 contingencies outside of the ATC system are those that are part of MISO's annual P2.4 MTEP reliability assessment that include monitoring the ATC system.

Dynamic analyses

Angular stability analysis

No P2.4 angular stability contingencies in the ATC system are included in the generator-centric lists. Angular instability is more severe for the BES generator under study for corresponding P4.6 contingencies because they include delayed clearing due to a stuck Bus-tie-Breaker. Therefore, each applicable P2.4 contingency is covered by a corresponding P4.6 contingency.

Voltage stability analysis

Load loss allowed – The P2.4 voltage stability contingencies in the ATC system that are included in study area-centric lists are those for which steady state screening studies indicate the potential for voltage instability.

The P2.4 dynamic contingencies outside of the ATC system are those that are part of MISO's annual P2.4 MTEP reliability assessment that include monitoring the ATC system.

P3 (Multiple Contingency - Loss of a generating unit followed by system adjustments and a subsequent fault on another Element)

These contingencies represent an outage of a BES generating unit followed by system adjustments and a subsequent three phase fault on one of five different types of BES Elements with normal clearing. The subsequent outages due to a fault on a generator, line, transformer, reactive device, or DC pole circuit are event-based contingencies based on the P1 contingency lists. All P3 contingencies are 'no load loss allowed' contingencies.

P3.1 (loss of a generator followed by System adjustments and then loss of another generator due to a three phase fault with normal clearing)

These contingencies represent an outage of a BES generating unit followed by system adjustments and a subsequent three phase fault on another BES generating unit with normal clearing. Since these contingencies are combinations of two BES generators, each combination pair is listed only once to avoid redundant analysis. It is assumed that for steady state analysis, the sequence of the generator outages with intervening system adjustments in each contingency pair produces the same final result. These outages are all event-based contingencies. All of P3.1 contingencies are 'no load loss allowed' contingencies.

Steady state analyses

Every year ATC generates a select list of applicable P3.1 contingencies, which include the following pairs of generators.

- The largest BES generator at each ATC substation paired with the largest BES generator at every other ATC substation.
- The largest BES generator at each ATC substation paired with the second largest BES generator at the same ATC substation.
- Largest generator is based on MW output capability.

These contingencies represent the pairing of the largest generation units on the ATC system. Therefore, ATC believes that these combinations of generator outages are the more severe P3.1 contingencies.

The P3.1 contingencies outside of the ATC system are those that are part of MISO's annual P3.1 MTEP reliability assessment that include monitoring the ATC system.

Dynamic analyses

Angular stability analysis

No P3.1 angular stability contingencies in the ATC system are included in the generator-centric lists. Angular stability is less severe for a BES generator under study than a P1.1 contingency because the first outage of any other BES generator followed by system adjustments is a more stable pre-P1.1 contingency condition and P1.2 or P1.3 contingencies are more stable than P1.1 contingencies. Therefore, each applicable P3.1 contingency is covered by a corresponding P1.2 or P1.3 contingency.

Voltage stability analysis

The P3.1 voltage stability contingencies in the ATC system that are included in study area-centric lists are those for which steady state screening studies indicate the potential for voltage instability.

The P3.1 dynamic contingencies outside of the ATC system are those that are part of MISO's annual P3.1 MTEP reliability assessment that include monitoring the ATC system.

P3.2 (loss of a generator followed by System adjustments and then loss of a transmission circuit due to a three phase fault with normal clearing)

These contingencies represent an outage of a BES generating unit followed by system adjustments and a subsequent three phase fault on a BES transmission line circuit with normal clearing. The line circuit outages are event-based contingencies. All of P3.2 contingencies are 'no load loss allowed' contingencies.

Steady state analyses

Every year ATC generates a select list of applicable P3.2 contingencies, which include the largest BES generator unit at each ATC substation paired with each BES line circuit in the ATC system.

These contingencies represent the pairing of the largest generation units on the ATC system. Therefore, ATC believes that these combinations of generator outages with any other BES line in the ATC system are the more severe P3.2 contingencies.

The P3.2 contingencies outside of the ATC system are those that are part of MISO's annual P3.2 MTEP reliability assessment that include monitoring the ATC system.

Dynamic Analyses

Angular stability analysis

No P3.2 angular stability contingencies in the ATC system are included in the generator-centric lists. Angular stability is less severe for a BES generator under study than a P1.2 contingency because the first outage of any other BES generator followed by system adjustments is a more stable pre-P1.2 contingency condition. Therefore, each applicable P3.2 contingency is covered by a corresponding P1.2 contingency.

Voltage stability analysis

The P3.2 voltage stability contingencies in the ATC system that are included in study area-centric lists are those for which steady state screening studies indicate the potential for voltage instability.

The P3.2 dynamic contingencies outside of the ATC system are those that are part of MISO's annual P3.2 MTEP reliability assessment that include monitoring the ATC system.

P3.3 (loss of a generator followed by System adjustments and then loss of a transformer due to a three phase fault with normal clearing)

These contingencies represent an outage of a BES generating unit followed by system adjustments and a subsequent three phase fault on a BES transformer circuit with normal clearing. The transformer circuit outages are event-based contingencies. All of P3.3 contingencies are 'no load loss allowed' contingencies.

Steady state analyses

Every year ATC generates a select list of applicable P3.3 contingencies, which include the largest BES generator unit at each ATC substation paired with each BES transformer circuit in the ATC system.

These contingencies represent the pairing of the largest generation units on the ATC system. Therefore, ATC believes that these combinations of generator outages with any other BES transformer in the ATC system are the more severe P3.3 contingencies.

The P3.3 contingencies outside of the ATC system are those that are part of MISO's annual P3.3 MTEP reliability assessment that include monitoring the ATC system.

Dynamic Analyses*Angular stability analysis*

No P3.3 angular stability contingencies in the ATC system are included in the generator-centric lists. Angular stability is less severe for a BES generator under study than a P1.3 contingency because the first outage of any other BES generator followed by system adjustments is a more stable pre-P1.3 contingency condition. Therefore, each applicable P3.3 contingency is covered by a corresponding P1.3 contingency.

Voltage stability analysis

The P3.3 voltage stability contingencies in the ATC system that are included in study area-centric lists are those for which steady state screening studies indicate the potential for voltage instability.

The P3.3 dynamic contingencies outside of the ATC system are those that are part of MISO's annual P3.3 MTEP reliability assessment that include monitoring the ATC system.

P3.4 (loss of a generator followed by System adjustments and then loss of a shunt device due to a three phase fault with normal clearing)

These contingencies represent an outage of a BES generating unit followed by system adjustments and a subsequent three phase fault on a BES shunt device with normal clearing. Multiple shunt device outages are modeled as a single event if they are connected to the bus through the same breaker. These outages are all event-based contingencies. All of P3.4 contingencies are 'no load loss allowed' contingencies.

Steady state analyses

Every year ATC generates a select list of applicable P3.4 contingencies, which include the largest BES generator unit at each ATC substation paired with each BES shunt device in the ATC system.

These contingencies represent the pairing of the largest generation units on the ATC system. Therefore, ATC believes that these combinations of generator outages with any other BES shunt device in the ATC system are the more severe P3.4 contingencies.

The P3.4 contingencies outside of the ATC system are those that are part of MISO's annual P3.4 MTEP reliability assessment that include monitoring the ATC system.

Dynamic Analyses*Angular stability analysis*

No P3.4 angular stability contingencies in the ATC system are included in the generator-centric lists. Angular stability is less severe for a BES generator under study than a P1.4 contingency because the first outage of any other BES generator followed by system adjustments is a more stable pre-P1.4 contingency condition and P1.2 or P1.3 contingencies are more severe than P1.4 contingencies. Therefore, each applicable P3.4 contingency is covered by a corresponding P1.2 or P1.3 contingency.

Voltage stability analysis

The P3.4 voltage stability contingencies in the ATC system that are included in study area-centric lists are those for which steady state screening studies indicate the potential for voltage instability.

The P3.4 dynamic contingencies outside of the ATC system are those that are part of MISO's annual P3.4 MTEP reliability assessment that include monitoring the ATC system.

P3.5 (loss of a generator followed by system adjustments and then loss of a single pole of a DC line due to a single line-to-ground fault with normal clearing)

These contingencies represent an outage of a BES generating unit followed by system adjustments and a subsequent SLG fault on a single pole of a BES DC line with normal (primary) clearing. The pole outages are event-based contingencies. All of P3.5 contingencies are 'no load loss allowed' contingencies.

The Mackinac back-to-back HVDC facility does not apply to this category because ATC interprets that there is no DC transmission line within the HVDC building (see Appendix).

Steady state analyses

There are presently no DC lines in the ATC system. Therefore, there are no P3.5 contingencies within the ATC system.

The P3.5 contingencies outside of the ATC system are those that are part of MISO's annual P3.5 MTEP reliability assessment that include monitoring the ATC system.

Dynamic analyses

Angular and Voltage stability analysis

There are presently no DC lines in the ATC system. Therefore, there are no P3.5 dynamic contingencies in the generator-centric or study area-centric lists.

The P3.5 dynamic contingencies outside of the ATC system are those that are part of MISO's annual P3.5 MTEP reliability assessment that include monitoring the ATC system.

P4 (Multiple Contingency - Loss of multiple Elements due to a fault on an Element plus a stuck breaker)

These contingencies represent a single-line-to-ground (SLG) fault on one of five different types of BES elements (generator, line, transformer, shunt device, or single pole of a DC line) where one of the fault clearing breakers fails to open. These outages are all event-based contingencies with the initial condition of a normal system.

P4.1 (loss multiple elements due to a SLG fault on a generator with delayed clearing due to a stuck breaker)

These contingencies represent a single-line-to-ground (SLG) fault on a BES generator terminal where one of the non-Bus-tie Breakers fails to open, which results in delayed clearing.

Steady state analyses

No load loss allowed - the applicable P4.1 contingencies for the ATC system outage the same elements as the corresponding P2.2 and P2.3 contingencies. The comprehensive list of 'no load loss allowed' P2.2 and P2.3 contingencies are generated every year.

Load loss allowed - the applicable P4.1 contingencies for the ATC system outage the same elements as the corresponding list of P2.2 and P2.3 contingencies. The comprehensive list of 'load loss allowed' P2.2 and P2.3 contingencies is generated every year from the current version of the master contingency list. The P4.1 contingencies outside of the ATC system are those that are part of MISO's annual P4.1 MTEP reliability assessment that include monitoring the ATC system.

Dynamic analyses

Angular stability analysis

No load loss allowed – No P4.1 EHV angular stability contingencies in the ATC system are included in the generator-centric lists. The BES generator connected to the ATC system that is under study is more stable when a BES generating unit is outaged after the delayed clearing due to a stuck breaker.

Load loss allowed - The P4.1 HV angular stability contingencies are treated the same as the EHV contingencies.

Voltage stability analysis

No load loss allowed – The P4.1 EHV voltage stability contingencies in the ATC system that are included in study area-centric lists are those for which steady state screening studies indicate the potential for voltage instability.

Load loss allowed - The P4.1 HV voltage stability contingencies in the ATC system that are included in study area-centric lists are those for which steady state screening studies indicate the potential for voltage instability.

The P4.1 dynamic contingencies outside of the ATC system are those that are part of MISO's annual P4.1 MTEP reliability assessment that include monitoring the ATC system.

P4.2 (loss of a transmission circuit due to a SLG fault with delayed clearing due to a stuck breaker)

These contingencies represent a single-line-to-ground (SLG) fault on any BES line terminals where any one of the non-Bus-tie Breakers fails to open, which results in delayed clearing.

Steady state analyses

No load loss allowed - the applicable P4.2 contingencies for the ATC system outage the same elements as the corresponding list of P2.3 contingencies. The comprehensive list of 'no load loss allowed' P2.3 contingencies is generated every year.

Load loss allowed - the applicable P4.2 contingencies for the ATC system outage the same elements as the corresponding list of P2.3 contingencies. The comprehensive list of 'load loss allowed' P2.3 contingencies is generated every year from the current version of the master contingency list. The P4.2 contingencies outside of the ATC system are those that are part of MISO's annual P4.2 MTEP reliability assessment that include monitoring the ATC system.

Dynamic analyses

Angular stability analysis

No load loss allowed – the P4.2 EHV angular stability contingencies in the ATC system are included in generator-centric lists when comparable S-E2.b contingencies result in angular instability. Angular instability is more severe for a BES generator connected to the ATC system that is under study for S-E2.b contingencies because the application of a three phase fault is more severe than a single line to ground fault. Therefore, each applicable P4.2 contingency can be covered by a corresponding S-E2.b contingency. The more severe P4.2 contingencies are those that are electrically closer to the study generator and removes stronger ties to the surrounding BES network, but do not trip any other generators.

Load loss allowed – The P4.2 HV angular stability contingencies are treated the same as the EHV contingencies.

Voltage stability analysis

No load loss allowed - The P4.2 EHV voltage stability contingencies in the ATC system that are included in study area-centric lists are those for which steady state screening studies indicate the potential for voltage instability.

Load loss allowed – The P4.2 HV voltage stability contingencies in the ATC system that are included in study area-centric lists are those for which steady state screening studies indicate the potential for voltage instability. The P4.2 dynamic contingencies outside of the ATC system are those that are part of MISO's annual P4.2 MTEP reliability assessment that include monitoring the ATC system.

P4.3 (loss of a transformer due to a SLG fault with delayed clearing due to a stuck breaker)

These contingencies represent a single-line-to-ground (SLG) fault on any BES transformer terminals where any one of the non-Bus-tie Breakers fails to open, which results in delayed clearing.

Steady state analyses

No load loss allowed - the applicable P4.3 contingencies for the ATC system outage the same elements as the corresponding list of P2.3 contingencies. The comprehensive list of 'no load loss allowed' P2.3 contingencies is generated every year.

Load loss allowed - the applicable P4.3 contingencies for the ATC system outage the same elements as the corresponding list of P2.3 contingencies. The comprehensive list of 'load loss allowed' P2.3 contingencies is generated every year from the current version of the master contingency list. The P4.3 contingencies outside of the ATC system are those that are part of MISO's annual P4.3 MTEP reliability assessment that include monitoring the ATC system.

Dynamic analyses

Angular stability analysis

No load loss allowed - The P4.3 EHV angular stability contingencies in the ATC system are included in the generator-centric lists when comparable S-E2.c contingencies result in angular instability. Angular instability is more severe for a BES generator connected to the ATC system that is under study for S-E2.c contingencies because the application of a three phase fault is more severe than a single line to ground fault. Therefore, each applicable P4.3 contingency is covered by a corresponding S-E2.c contingency. The more severe P4.3 contingencies are those that are electrically closer to the study generator and removes stronger ties to the surrounding BES network without outaging any other generators.

Load loss allowed – The P4.3 HV angular stability contingencies are treated the same as the EHV contingencies.

Voltage stability analysis

No load loss allowed – The P4.3 EHV voltage stability contingencies in the ATC system that are included in study area-centric lists are those for which steady state screening studies indicate the potential for voltage instability.

Load loss allowed – The P4.3 HV voltage stability contingencies in the ATC system that are included in study area-centric lists are those for which steady state screening studies indicate the potential for voltage instability.

The P4.3 dynamic contingencies outside of the ATC system are those that are part of MISO's annual P4.3 MTEP reliability assessment that include monitoring the ATC system.

P4.4 (loss of a shunt device due to a SLG fault with delayed clearing due to a stuck breaker)

These contingencies represent a single-line-to-ground (SLG) fault on any BES shunt device terminals where any one of the non-Bus-tie Breakers fails to open, which results in delayed clearing.

Steady state analyses

No load loss allowed - the applicable P4.4 contingencies for the ATC system outage the same elements as the corresponding list of P2.3 contingencies. The comprehensive list of 'no load loss allowed' P2.3 contingencies is generated every year.

Load loss allowed - the applicable P4.4 contingencies for the ATC system outage the same elements as the corresponding list of P2.3 contingencies. The comprehensive list of 'load loss allowed' P2.3 contingencies is generated every year from the current version of the master contingency list. The P4.4 contingencies outside of the ATC system are those that are part of MISO's annual P4.4 MTEP reliability assessment that include monitoring the ATC system.

Dynamic analyses

Angular stability analysis

No load loss allowed – No P4.4 EHV angular stability contingencies in the ATC system are included in the generator-centric lists. The P4.4 shunt device contingencies will not cause more severe angular instability of the BES generator under study than a P4.2, P4.3, P4.5 or P4.6 contingency because shunt device contingencies do not remove any ties to the surrounding BES network after the contingency event. Therefore, each applicable P4.4 contingency is covered by a corresponding P4.2, P4.3, P4.5, or P4.6 contingency.

Load loss allowed – The P4.4 HV angular stability contingencies are treated the same as the EHV contingencies.

Voltage stability analysis

No load loss allowed – The P4.4 EHV voltage stability contingencies in the ATC system that are included in study area-centric lists are those for which steady state screening studies indicate the potential for voltage instability.

Load loss allowed – The P4.4 HV voltage stability contingencies in the ATC system that are included in study area-centric lists are those for which steady state screening studies indicate the potential for voltage instability.

The P4.4 dynamic contingencies outside of the ATC system are those that are part of MISO's annual P4.4 MTEP reliability assessment that include monitoring the ATC system.

P4.5 (loss of multiple elements due to a stuck non-bus-tie breaker attempting to clear a SLG fault on a bus section)

These contingencies represent a single-line-to-ground (SLG) fault on any BES bus section terminals where any one of the non-Bus-tie Breakers fails to open, which results in delayed clearing.

Steady state analyses

No load loss allowed - the applicable P4.5 contingencies for the ATC system outage the same elements as the corresponding list of P2.3 contingencies. The comprehensive list of 'no load loss allowed' P2.3 contingencies is generated every year.

Load loss allowed - the applicable P4.5 contingencies for the ATC system outage the same elements as the corresponding list of P2.3 contingencies. The comprehensive list of 'load loss allowed' P2.3 contingencies is generated every year from the current version of the master contingency list. The P4.5 contingencies outside of the ATC system are those that are part of MISO's annual P4.5 MTEP reliability assessment that include monitoring the ATC system.

Dynamic analyses

Angular stability analysis

No load loss allowed - The P4.5 EHV angular stability contingencies in the ATC system are included in the generator-centric lists when comparable S-E2.d contingencies result in angular instability. Angular instability is more severe for a BES generator connected to the ATC system that is under study for S-E2.d contingencies because the application of a three phase fault is more severe than a single line to ground fault. Therefore, each applicable P4.5 contingency is covered by a corresponding S-E2.d contingency. The more severe P4.5 contingencies are those applied at EHV bus sections that are closer to the study generator and removes stronger ties to the surrounding BES network without outaging any other generators.

Load loss allowed – The P4.5 HV angular stability contingencies are treated the same as the EHV contingencies.

Voltage stability analysis

No load loss allowed – The P4.5 EHV voltage stability contingencies in the ATC system that are included in study area-centric lists are those for which steady state screening studies indicate the potential for voltage instability.

Load loss allowed – The P4.5 HV voltage stability contingencies in the ATC system that are included in study area-centric lists are those for which steady state screening studies indicate the potential for voltage instability.

The P4.5 dynamic contingencies outside of the ATC system are those that are part of MISO's annual P4.5 MTEP reliability assessment that include monitoring the ATC system.

P4.6 (loss of multiple elements due to a Bus-tie Breaker attempting to clear a SLG fault on one of the associated buses)

These contingencies represent a single-line-to-ground (SLG) fault on any BES buses associated a Bus-tie breaker where any one of the non-Bus-tie Breakers fails to open, which results in delayed clearing. All of P4.6 contingencies are 'load loss allowed' contingencies.

Steady state analyses

Load loss allowed - the applicable P4.6 contingencies for the ATC system outage the same elements as the corresponding list of P2.4 contingencies. The comprehensive list of 'load loss allowed' P2.4 contingencies is generated every year from the current version of the master contingency list. The P4.6 contingencies outside of the ATC system are those that are part of MISO's annual P4.6 MTEP reliability assessment that include monitoring the ATC system.

Dynamic analyses

Angular stability analysis

Load loss allowed - the P4.6 angular stability contingencies in the ATC system are included in the generator-centric lists when comparable S-E2.d contingencies result in angular instability. Angular instability is more severe for a BES generator connected to the ATC system that is under study for S-E2.d contingencies because it application of a three phase fault is more severe than a single line to ground fault. Therefore, each applicable P4.6 contingency is covered by a corresponding S-E2.d contingency. Therefore, P4.6 contingencies can be covered by corresponding S-E2.d contingencies. The more severe P4.6 contingencies are those that are closer to generating units and outage significant outlets for generating units, but do not trip any of the units.

Voltage stability analysis

Load loss allowed – The P4.6 voltage stability contingencies in the ATC system that are included in study area-centric lists are those for which steady state screening studies indicate the potential for voltage instability.

The P4.6 dynamic contingencies outside of the ATC system are those that are part of MISO's annual P4.6 MTEP reliability assessment that include monitoring the ATC system.

P5 (Multiple Contingency - Loss of multiple Elements due to delayed fault clearing due to failure of a non-redundant relay)

These contingencies represent a single-line-to-ground (SLG) fault on one of five different types of BES elements (generator, line, transformer, shunt device, or bus section) with delayed fault clearing due to the failure of a non-redundant relay protecting the faulted element to operate as designed. These outages are all event-based contingencies with the initial condition of a normal system.

The relay functions or types that are considered include those specified in Note 13 of Table 1 of the TPL-001-4 standard. These relay functions are pilot (#85), distance (#21), differential (#87), current (#50, #51, and #67), voltage (#27 & #59), directional (#32, & #67), and tripping (#86, & #94).

P5.1 (loss of a generator due to SLG fault with delayed clearing due to failure of a non-redundant relay)

These contingencies represent a single-line-to-ground (SLG) fault on a BES generator terminal with delayed fault clearing due to the failure of a non-redundant relay protecting the faulted element to operate as designed.

Steady state analyses

No load loss allowed - there are no applicable P5.1 contingencies for the ATC system because all generators that are interconnected at the EHV level have redundant protective relays.

Load loss allowed - there are no applicable P5.1 contingencies for the ATC system because all generators that are interconnected at the HV level have redundant protective relays.

ATC practice is to include redundant relays for any new generator interconnected to the ATC system, thus no new P5.1 contingencies would be expected in the future.

The P5.1 contingencies outside of the ATC system are those that are part of MISO's annual P5.1 MTEP reliability assessment that include monitoring the ATC system.

Dynamic analyses

Angular and Voltage stability analysis

No load loss allowed - there are no applicable P5.1 dynamic contingencies for the ATC system because all generators that are interconnected at the EHV level have redundant protective relays.

Load loss allowed - there are no applicable P5.1 dynamic contingencies for the ATC system because all generators that are interconnected at the HV level have redundant protective relays.

The P5.1 dynamic contingencies outside of the ATC system are those that are part of MISO's annual P5.1 MTEP reliability assessment that include monitoring the ATC system.

P5.2 (loss of a transmission circuit due to SLG fault with delayed clearing due to failure of a non-redundant relay)

These contingencies represent a single-line-to-ground (SLG) fault on any BES line terminals with delayed fault clearing due to the failure of a non-redundant relay protecting the faulted element to operate as designed.

Steady state analyses

No load loss allowed - there are no applicable P5.2 contingencies for the ATC system because all EHV transmission circuits have redundant protective relays.

Load loss allowed - there are no applicable P5.2 contingencies for the ATC system because all HV transmission circuits have redundant protective relays.

ATC protection standards require redundant relays for any new transmission line added to the ATC system, thus no new P5.2 contingencies would be expected in the future.

The P5.2 contingencies outside of the ATC system are those that are part of MISO's annual P5.2 MTEP reliability assessment that include monitoring the ATC system.

Dynamic analyses

Angular and Voltage stability analysis

No load loss allowed - there are no applicable P5.2 dynamic contingencies for the ATC system because all EHV transmission circuits have redundant protective relays.

Load loss allowed - there are no applicable P5.2 dynamic contingencies for the ATC system because all HV transmission circuits have redundant protective relays.

The P5.2 dynamic contingencies outside of the ATC system are those that are part of MISO's annual P5.2 MTEP reliability assessment that include monitoring the ATC system.

P5.3 (loss of a transmission transformer due to SLG fault with delayed clearing due to failure of a non-redundant relay)

These contingencies represent a single-line-to-ground (SLG) fault on any BES transformer terminals with delayed fault clearing due to the failure of a non-redundant relay protecting the faulted element to operate as designed.

Steady state analyses

No load loss allowed - there are no applicable P5.3 contingencies for the ATC system because there are no EHV transformers in the ATC system.

Load loss allowed - there is presently only one applicable P5.3 contingency for the ATC system, the Oak Creek 230 kV/138 kV transformer, because all other transmission transformers have redundant relaying. This contingency outages the same elements as the two Oak Creek 138 kV bus P5.5 contingencies, so the Oak Creek 230 kV/138 kV transformer P5.3 contingency is covered by the Oak Creek 138 kV bus P5.5 contingencies.

ATC protection standards require redundant relays for any new transformers added to the ATC system, thus no new P5.3 contingencies would be expected in the future.

The P5.3 contingencies outside of the ATC system are those that are part of MISO's annual P5.3 MTEP reliability assessment that include monitoring the ATC system.

Dynamic analyses

Angular and Voltage stability analysis

No load loss allowed - there are no applicable P5.3 dynamic contingencies for the ATC system because there are no EHV transformers in the ATC system.

Load loss allowed - there is presently only one applicable P5.3 dynamic contingency for the ATC system, the Oak Creek 230 kV/138 kV transformer, because other applicable contingencies have redundant relaying. The Oak Creek transformer P5.3 contingency is included in generator stability

studies for the Oak Creek 138 kV connected generator, Oak Creek 230 kV connected generators, or Elm Road 345 kV connected generators. Because of the abundance of generation and network interconnectivity in the local area, there is presently no need for voltage stability studies in this local area.

The P5.3 dynamic contingencies outside of the ATC system are those that are part of MISO's annual P5.3 MTEP reliability assessment that include monitoring the ATC system.

P5.4 (loss of a shunt device due to a SLG fault due to a relay failure)

These contingencies represent a single-line-to-ground (SLG) fault on any BES shunt device terminals with delayed fault clearing due to the failure of a non-redundant relay protecting the faulted element to operate as designed.

Steady state analyses

No load loss allowed - there are no applicable P5.4 contingencies for the ATC system because all EHV shunt devices have redundant protective relays.

Load loss allowed - there are no applicable P5.4 contingencies for the ATC system because all HV shunt devices have redundant protective relays.

ATC protection standards require redundant relays for any new shunt device added to the ATC system, thus no new P5.4 contingencies would be expected in the future.

The P5.4 contingencies outside of the ATC system are those that are part of MISO's annual P5.4 MTEP reliability assessment that include monitoring the ATC system.

Dynamic analyses

Angular and Voltage stability analysis

No load loss allowed - there are no applicable P5.4 dynamic contingencies for the ATC system because all EHV shunt devices have redundant protective relays.

Load loss allowed - there are no applicable P5.4 dynamic contingencies for the ATC system because all HV shunt devices have redundant protective relays.

The P5.4 dynamic contingencies outside of the ATC system are those that are part of MISO's annual P5.4 MTEP reliability assessment that include monitoring the ATC system.

P5.5 (loss of multiple elements due to a SLG fault on a bus section due to a relay failure)

These contingencies represent a single-line-to-ground (SLG) fault on any BES bus section with delayed fault clearing due to the failure of a non-redundant relay protecting the faulted element to operate as designed.

Steady state analyses

No load loss allowed - the only EHV buses for the ATC system that do not have redundant relays are covered by four applicable P5.5 contingencies – Arcadian 345 kV, Granville 345 kV, Point Beach 345 kV, and Racine 345 kV buses.

Load loss allowed - the applicable P5.5 contingencies for the ATC system are presently limited to select 115 kV and 138 kV straight buses in the ATC system because some bus configurations have redundant relays. These contingencies are covered by the corresponding list of P2.3 (for non-segmented straight bus) or P2.4 (for segmented straight bus) contingencies. The comprehensive list of 'load loss allowed'

P2.3 and P2.4 contingencies is generated every year from the current version of the master contingency list.

ATC protection standards require redundant relays for any new transmission bus added to the ATC system, thus no new P5.5 contingencies would be expected in the future. In addition, when redundant relays are added on existing transmission buses, these buses will be removed from the contingency list.

The P5.5 contingencies outside of the ATC system are those that are part of MISO's annual P5.5 MTEP reliability assessment that include monitoring the ATC system.

Dynamic analyses

Angular stability analysis

No load loss allowed - the only EHV buses for the ATC system that do not have redundant relays are covered by four applicable P5.5 contingencies – Arcadian 345 kV, Granville 345 kV, Point Beach 345 kV, and Racine 345 kV buses. The P5.5 EHV angular stability contingencies in the ATC system are included in study area-centric list when comparable S-E2.d contingencies result in angular instability. Angular instability is more severe for S-E2.d contingencies because the application of a three phase fault is more severe than a line to ground fault. Therefore, each applicable P5.5 contingency is covered by a corresponding S-E2.d contingency. The more severe P5.5 contingencies are those applied at the EHV bus sections that are electrically closer to study generator and remove more and stronger ties to the surrounding BES network, but without outaging any other generators.

Load loss allowed - the applicable P5.5 contingencies for the ATC system are presently limited to select 115 kV and 138 kV straight buses in the ATC system because some bus configurations do not have redundant relays. The selected P5.5 HV contingencies are treated the same as the EHV contingencies.

Voltage stability analysis

No load loss allowed - the only EHV buses for the ATC system that do not have redundant relays are covered by four applicable P5.5 contingencies – Arcadian 345 kV, Granville 345 kV, Point Beach 345 kV, and Racine 345 kV buses. The P5.5 voltage stability contingencies in the ATC system that are included in study area-centric lists are those for which steady state screening studies indicate the potential for voltage instability.

Load loss allowed - the applicable P5.5 contingencies for the ATC system are presently limited to select 115 kV and 138 kV straight buses in the ATC system because some bus configurations do not have redundant relays. The selected P5.5 HV contingencies are treated the same as the EHV contingencies.

The P5.5 dynamic contingencies outside of the ATC system are those that are part of MISO's annual P5.5 MTEP reliability assessment that include monitoring the ATC system.

P6 (Multiple Contingency – Loss of an Element followed by System adjustments and a subsequent fault on another Element)

These contingencies represent the outage of one of four kinds of transmission Elements (line, transformer, shunt device, or single pole of a DC line) followed by system adjustments and a subsequent fault on any one of the same four types of transmission Elements with normal clearing. The subsequent line, transformer, shunt devices, and DC pole circuit outages are all event-based contingencies.

P6.1 (Loss of a transmission circuit followed by System adjustments and a subsequent fault on a specific types of Elements)

These contingencies represent an outage of a BES line (P1.2) followed by System adjustments and a subsequent three-phase fault with normal clearing on either a:

- BES line (P1.2),
- BES transformer (P1.3),
- BES shunt device (P1.4), or
- Single pole of a BES DC line (P1.5).

Steady state analyses

Load loss allowed - every year ATC generates a comprehensive list of applicable P6.1 contingencies from the current version of the master contingency list. The P6.1 contingencies outside of the ATC system are those that are part of MISO's annual P6.1 MTEP reliability assessment that include monitoring the ATC system.

Dynamic analyses

Angular stability analysis

Load loss allowed - The P6.1 angular stability contingencies in the ATC system are included in the generator-centric lists when any comparable S-E1 contingencies resulted in angular instability. Angular instability for a S-E1 contingency is the same or more severe than the comparable P6.1 contingency because the S-E1 contingency does not allow system adjustment after the initial outage.

When a P6.1 contingency is included in the list, the first outage is a line (P1.2) and the second outage is a line (P1.2), transformer (P1.3) or shunt (P1.4).

Voltage stability analysis

Load loss allowed – The P6.1 voltage stability contingencies in the ATC system that are included in study area-centric lists are those for which steady state screening studies indicate the potential for voltage instability.

When a P6.1 contingency is included in the list, the first outage is a line (P1.2) and the second outage is a line (P1.2), transformer (P1.3) or shunt (P1.4).

The P6.1 dynamic contingencies outside of the ATC system are those that are part of MISO's annual P6.1 MTEP reliability assessment that include monitoring the ATC system.

P6.2 (Loss of a transmission transformer followed by System adjustments and a subsequent fault on a specific Element)

These contingencies represent an outage of BES transformer (P1.3) followed by System adjustments and a subsequent three-phase fault with normal clearing on either a:

- BES line (P1.2),
- BES transformer (P1.3),
- BES shunt device (P1.4), or
- Single pole of a BES DC line (P1.5).

Steady state analyses

Load loss allowed - every year ATC generates a comprehensive list of applicable P6.2 contingencies from the current version of the master contingency list. The P6.2 contingencies outside of the ATC system are those that are part of MISO's annual P6.2 MTEP reliability assessment that include monitoring the ATC system.

Dynamic Analyses

Angular stability analysis

Load loss allowed - The P6.2 angular stability contingencies in the ATC system are included in the generator-centric lists when comparable S-E1 contingencies result in angular instability. Angular instability is the same or be more severe for a BES generator connected to the ATC system that is under study for S-E1 contingencies because they do not allow system adjustment after the initial outage. Therefore, each applicable P6.2 contingency is covered by a corresponding S-E1 contingency. The more severe P6.2 contingencies are those where the first outage is BES transformer that is a strong tie from the study generator area to the surrounding BES network followed by system adjustment and and the second outage is a P1.2, P1.3, or P1.4 fault electrically closer to the study generator on another strong tie (BES line or BES transformer) to the surrounding BES network, but without outaging any other generators.

Voltage stability analysis

Load loss allowed – The P6.2 voltage stability contingencies in the ATC system that are included in study area-centric lists are those for which steady state screening studies indicate the potential for voltage instability.

When a P6.2 contingency is included in the list, the first outage is a transformer (P1.3) and the second outage is a line (P1.2), transformer (P1.3) or shunt (P1.4).

The P6.2 dynamic contingencies outside of the ATC system are those that are part of MISO's annual P6.2 MTEP reliability assessment that include monitoring the ATC system.

P6.3 (Loss of a shunt device followed by System adjustments and a subsequent fault on a specific Element)

These contingencies represent an outage of BES shunt device (P1.4) followed by System adjustments and a subsequent three-phase fault with normal clearing on either a:

- BES line (P1.2),
- BES transformer (P1.3),
- BES shunt device (P1.4), or
- Single pole of a BES DC line (P1.5).

Steady state analyses

Load loss allowed - every year ATC generates a comprehensive list of applicable P6.3 contingencies from the current version of the master contingency list. The P6.3 contingencies outside of the ATC system are those that are part of MISO's annual P6.3 MTEP reliability assessment that include monitoring the ATC system.

Dynamic analyses*Angular stability analysis*

Load loss allowed - No P6.3 angular stability contingencies in the ATC system are included in the generator-centric lists because the generator under study is more stable for a P6.3 contingency than a P6.1 or P6.2 contingency. Angular stability is the same or better for a BES generator connected to the ATC system that is under study for a P6.3 contingency because the first outage is a shunt device, which does not outage other generation near the study generator or remove a tie from the study generator area to the surrounding BES system. However, the first outage of a P6.1 or P6.2 contingency can remove a strong tie (BES line or BES transformer) to the surrounding BES network and be followed by the same element outage as a P6.3 contingency.

Voltage stability analysis

Load loss allowed – The P6.3 voltage stability contingencies in the ATC system that are included in study area-centric lists are those for which steady state screening studies indicate the potential for voltage instability.

When a P6.3 contingency is included in the list, the first outage is a shunt device (P1.4) and the second outage is a line (P1.2), transformer (P1.3) or shunt (P1.4).

The P6.3 dynamic contingencies outside of the ATC system are those that are part of MISO's annual P6.3 MTEP reliability assessment that include monitoring the ATC system.

P6.4 (Loss of a single pole of a DC line followed by System adjustments and a subsequent fault on a specific Element)

These contingencies represent an outage of single pole of BES DC line (P1.5) followed by System adjustments and a subsequent three-phase fault with normal clearing on either a:

- BES line (P1.2),
- BES transformer (P1.3),
- BES shunt device (P1.4), or
- Single pole of a BES DC line (P1.5).

The Mackinac back-to-back HVDC facility does not apply to the single pole of BES DC line because ATC interprets that there is no DC transmission line within the HVDC building (see Appendix).

Steady state analyses

There are presently no BES DC lines in the ATC system. Therefore, there are no P6.3 steady state contingencies are included in the Contingency List for the ATC system.

The P6.4 contingencies outside of the ATC system are those that are part of MISO's annual P6.4 MTEP reliability assessment that include monitoring the ATC system.

Dynamic analyses

Angular and Voltage stability analysis

There are presently no DC lines in the ATC system. Therefore, there are no P6.4 dynamic contingencies in the generator-centric or study area-centric lists.

The P6.4 dynamic contingencies outside of the ATC system are those that are part of MISO's annual P6.4 MTEP reliability assessment that include monitoring the ATC system.

P7 (Multiple Contingency - Loss of any two circuits on a common structure)

These contingencies represent a single-line-to-ground (SLG) fault on any two BES circuits on a common structure with normal clearing. These outages are all event-based contingencies. ATC only selects P7 contingencies for circuits that share a common structure for a cumulative length of one mile or more.

P7.1 (loss of any two adjacent circuits on a common structure)

These contingencies represent a single-line-to-ground (SLG) fault with normal clearing on any two adjacent (vertically or horizontally) BES circuits on a common structure. The line outages are event-based contingencies with prior normal system conditions. All of P7.1 contingencies are 'load loss allowed' contingencies.

Steady state analyses

Load loss allowed - every year ATC generates a comprehensive list of applicable P7.1 contingencies from the current version of the master contingency list. The P7.1 contingencies outside of the ATC system are those that are part of MISO's annual P7.1 MTEP reliability assessment that include monitoring the ATC system.

Dynamic analyses

Angular stability analysis

Load loss allowed - The P7.1 angular stability contingencies in the ATC system are included in generator-centric lists for a BES generator connected to the ATC system under study when it is more severe than any other type of contingency. Angular instability is more severe for a BES generator that is under study for any contingencies that apply a SLG fault to the end of both BES lines that is electrically closer to the study generator or removes stronger ties from the study generator area to the surrounding BES network, but without outaging other generators.

Voltage stability analysis

Load loss allowed - The P7.1 voltage stability contingencies in the ATC system that are included in study area-centric lists are those for which steady state screening studies indicate the potential for voltage instability.

The P7.1 dynamic contingencies outside of the ATC system are those that are part of MISO's annual P7.1 MTEP reliability assessment that include monitoring the ATC system.

P7.2 (Loss of a bipolar DC line)

These contingencies represent a single-line-to-ground (SLG) fault with normal clearing on both poles of a bipolar BES DC line on a common structure. All of P7.2 contingencies are 'load loss allowed' contingencies.

Steady state analyses

There are presently no bipolar DC lines in the ATC system. Therefore, there are no P7.2 contingencies within the ATC system.

The P7.2 steady state contingencies outside of the ATC system are those that are part of MISO's annual P7.2 MTEP reliability assessment that include monitoring the ATC system.

Dynamic analyses

Angular and Voltage stability analysis

There are presently no bipolar BES DC lines in the ATC system. Therefore, there are no P7.2 dynamic contingencies in the generator-centric or study area-centric lists.

The P7.2 dynamic contingencies outside of the ATC system are those that are part of MISO's annual P7.2 MTEP reliability assessment that include monitoring the ATC system.

Section 3.2 Extreme Events Contingencies

Section 3.2.1 Steady State Analysis Extreme Event Contingencies

SS-E1 (Two overlapping singles without system adjustments)

These contingencies represent the outage of a single BES Element (generator, Transmission Circuit, single pole of a DC Line, shunt device, or transformer) followed by the outage of another single BES Element (generator, Transmission Circuit, single pole of a different DC Line, shunt device, or transformer) without System adjustments. The BES Element outages are event-based contingencies.

These contingencies simply represent the P3 and P6 contingencies without System adjustments.

The SS-E1 contingencies inside and outside of the ATC system are those that are part of MISO's annual SS-E1 contingencies list that it uses for the annual MTEP Planning Assessment. MISO creates the SS-E1 (two overlapping singles) contingencies from annual P1 single contingencies it receives from ATC and other entities outside of the ATC system.

SS-E2 (Local area events affecting the Transmission System)

There are five different contingencies that are considered in the SS-E2 group.

- a. Loss of a tower line with three or more circuits
- b. Loss of all Transmission lines on a common Right-of-Way
- c. Loss of a switching station or substation (loss of one voltage level plus transformers).
- d. Loss of all generating units at a generating station.
- e. Loss of a large Load or major Load center.

Every year ATC generates a comprehensive list of applicable ATC system SS-E2 contingencies from the previous version of the SS-E2 contingency list and provides this list to MISO.

The SS-E2 contingencies inside and outside of the ATC system are those that are part of MISO's annual SS-E2 contingencies list that it uses for the annual MTEP Planning Assessment.

SS-E2.a (loss of tower line with three or more circuits)

These contingencies represent an outage of a tower line (common structure) with three or more BES transmission circuits. All of the BES and non-BES transmission circuits on the tower line are outaged. The line outages are event-based contingencies with prior normal system conditions. Only the combinations of circuits that share a cumulative distance of one mile or more are listed.

SS-E2.b (all transmission lines in a common right-of-way)

ATC understands a "common right-of-way" to be either a single right-of-way or multiple rights-of-way physically adjacent to one another (i.e. no separation between rights-of-way).

These contingencies represent an outage on all the transmission circuits in a common Right-of-Way (ROW) that include at least two BES circuits. These contingencies include the outage of all BES and non-BES overhead and/or underground transmission circuits within a common right-of-way with normal clearing.

For ROWs with only two BES circuits, each applicable SS-E2.b contingency outages the same elements as the corresponding P6.1 or P7.1 contingency (without system adjustment) and does not have to be duplicated. For ROWs with three or more BES overhead circuits on a common tower line, each applicable SS-E2.b contingency outages the same elements as the corresponding SS-E2.a contingency and does not have to be duplicated.

All remaining SS-E2.b contingencies with three or more BES overhead or underground circuits that share a ROW are listed. The line outages are event-based contingencies with prior normal system conditions. Only the combinations of overhead and/or underground BES circuits that share a cumulative distance of one mile or more are listed.

SS-E2.c (loss of one voltage level plus transformers at a switching station or substation)

These contingencies represent an outage of all the elements at one BES voltage level plus the transformers connected to that voltage level with normal clearing at the switching station or substation. The common voltage bus outages are event-based contingencies with prior normal system conditions.

- For substations that are connected by only one BES transmission circuit and no shunt reactive device connected at the same voltage level, then each applicable SS-E2.c contingency outages the same elements as the corresponding P1.2 contingency and does not have to be duplicated.
- For substations that are connected by only two BES transmission circuits and no shunt reactive devices connected at the same voltage level, then each applicable SS-E2.c contingency outages the same elements as the corresponding P6.1 or P7.1 contingency and does not have to be duplicated.

All remaining SS-E2.c contingencies involving substations that are connected by three or more BES transmission circuits and/or BES generators are listed.

SS-E2.d (loss of all generating units at a generating station)

These contingencies represent an outage of all transmission generating units at the same generating station with normal clearing when there is at least one BES generating unit at the station. All of the BES generating units and any non-BES generating units at the generating station are outaged. The generating unit outages are event-based contingencies with prior normal system conditions. Refer to the Definition section for the definition of generating station.

For SS-E2.d contingencies within the ATC system, certain planning event contingencies cover some of the SS-E2.d contingencies and do not have to be repeated:

- For only one BES generating unit at a generating station, each applicable SS-E2.d contingency outages the same elements as the corresponding P1.1 contingency.
- For two BES generating units at a generating station, each applicable SS-E2.d contingency outages the same elements as the corresponding P3.1 contingency.
- For generating stations with any number of units that are connected through only one line circuit, each applicable SS-E2.d contingency outages the same elements as the corresponding P1.2 contingency.
- For generating stations with any number of units that are connected through only one transformer circuit (including a GSU transformer), each applicable SS-E2.d contingency outages the same elements as the corresponding P1.3 contingency.
- For generating stations with any number of units that are connected through any combination of two line circuits, two transformer circuits or a line and transformer circuit, each applicable SS-E2.d contingency outages the same elements as the corresponding P6.1 or P6.2 contingency.

All remaining SS-E2.d contingencies are listed.

SS-E2.e (loss of a large load or major load center)

These contingencies represent an outage of all the Load connected from a single BES substation with normal clearing. These outages are contingencies with prior normal system conditions.

ATC understands “a large load” to be a single load from a single substation of 300 MW or more and “a major load center” to be a group of loads from a single substation with an aggregate load of 300 MW or more. The 300 MW threshold is consistent with the NERC Category 2 event classification and the Department of Energy Electric Emergency Incident and Disturbance Report threshold.

There are presently no applicable large loads or major load centers in the ATC system.

SS-E3. Wide area events affecting the Transmission System based on System topology such as:

Every year ATC generates a comprehensive list of applicable ATC system SS-E3 contingencies from the previous version of the SS-E3 contingency list and provides this list to MISO.

The SS-E3 contingencies inside and outside of the ATC system are those that are part of MISO's annual SS-E3 contingencies list that it uses for the annual MTEP Planning Assessment.

SS-E3.a (loss of two generating stations)

These contingencies represent the outage of all BES and non-BES Facilities at two different BES generating stations due to any one of six different types of events. A BES generating station must contain at least one BES generating unit. The generating station outages are event-based contingencies with prior normal system conditions. Refer to the Definition section for the definition of generating station. When all the generating units at a generating station are composite generator models (generating unit, its auxiliary load and GSU transformer), then the combination of these models can be used to represent the loss of the entire generating station.

SS-E3.a.i (due to loss of a large gas pipeline)

These contingencies represent a loss of all the gas-fired generating units at two different generating stations due to a loss of a large common gas pipeline into a region or multiple regions that have significant gas-fired generation.

The consideration of the loss of a large gas pipeline will limit the applicable generating station contingency combinations to those that share a common gas pipeline. However, the SS-E3.a.v contingencies cover every possible two generation station contingency combination throughout the entire ATC area. Therefore, each applicable SS-E3.a.i contingency is covered by a corresponding SS-E3.a.v contingency and does not have to be duplicated.

SS-E3.a.ii (due to loss of the use of a large body of water as the cooling source for generation)

These contingencies represent a loss of all the generating units at two different BES generating stations due to a loss of the use of a large body of water as the cooling source for generation.

The consideration of the loss of a cooling source will limit the applicable generating station contingency combinations to those that share a common body of water. However, the SS-E3.a.v contingencies cover every possible two generation station contingency combination throughout the entire ATC area. Therefore, each applicable SS-E3.a.ii contingency is covered by a corresponding SS-E3.a.v contingency and does not have to be duplicated.

SS-E3.a.iii (due to wildfires)

These contingencies represent a loss of all the generating units at two different BES generating stations due to wildfires.

The consideration of wildfires will limit the applicable generating station contingency combinations to those in a specific local area. However, the SS-E3.a.v contingencies cover every possible two generation station contingency combination throughout the entire ATC area. Therefore, each applicable SS-E3.a.iii contingency is covered by a corresponding SS-E3.a.v contingency and does not have to be duplicated.

SS-E3.a.iv (ss) (due to severe weather)

These contingencies represent a loss of all the generating units at two different BES generating stations due to severe weather.

The consideration of severe weather will limit the applicable generating station contingency combinations to those in a specific local area. However, the SS-E3.a.v contingencies cover every possible two generation station contingency combination throughout the entire ATC area. Therefore, each applicable SS-E3.a.iv contingency is covered by a corresponding SS-E3.a.v contingency and does not have to be duplicated.

SS-E3.a.v (due to a successful cyber attack)

These contingencies represent the outage of all the generating units at two different BES generating stations due to a successful cyber attack.

ATC understands that a successful cyber attack outages all of the generating units at a generating station that can be tripped by one or more cyber assets. This will include all of the generating units at the station and depending on the interconnection bus configuration (e.g. straight, ring, breaker-and-a-half) may include the opening of breakers at the transmission substation.

SS-E3.a.vi (due to shutdown of a nuclear power plant(s) and related facilities for a day or more for common causes such as problems with similarly designed plants)

The consideration of the shutdown of a nuclear power plant will limit the applicable generating station contingency combinations to those which involve a nuclear power plant in the ATC system. However, the SS-E3.a.v contingencies cover every possible two generation station contingency combination throughout the entire ATC area. Therefore, each applicable SS-E3.a.vi contingency is covered by a corresponding SS-E3.a.v contingency and does not have to be duplicated.

SS-E3.b (Other events based upon operating experience that may result in wide area disturbances)

ATC understands this to mean that these contingencies represent any operating experience events that have resulted in the loss of 1000 MW or more of firm load. The 1000 MW threshold is the planning horizon IROL criteria for ATC and MISO.

ATC will consider operating experiences over the previous 10 years.

Section 3.2.2 Stability (Dynamics) Extreme Event Analysis

Each dynamic stability contingency is event-based (all Elements are cleared by a breaker or breaker-to-breaker operation).

The Stability contingencies events outside of the ATC system are those that are part of MISO's annual MTEP reliability assessment that include monitoring the ATC system.

Every year MISO considers what areas in its footprint need extreme event contingencies and asks ATC for Stability Extreme Event contingencies that are in the ATC system area. ATC then creates a list of Stability Extreme Event contingencies for the requested area and provides this a list to MISO.

The Stability Extreme Event contingencies inside and outside of the ATC system are those that are part of MISO's annual Stability Extreme Event contingencies list that it uses for the annual MTEP Planning Assessment.

S-E1. With an initial condition of a single generator, Transmission circuit, single pole of a DC line, shunt device, or transformer forced out of service, apply a 3Ø fault on another single generator,

Transmission circuit, single pole of a different DC line, shunt device, or transformer prior to System adjustments.

These contingencies represent the prior outage of one of five kinds of BES Elements:

- BES generator (P1.1)
- BES line (P1.2),
- BES transformer (P1.3),
- BES shunt device (P1.4), or
- Single pole of a BES DC line (P1.5).

Followed by a subsequent 3-phase fault with normal clearing for any other Elements in these contingency types prior to any System adjustments.

Angular stability analysis

The S-E1 angular stability contingencies included in a generator-centric list are those that will cause more severe angular instability for a BES generator connected to the ATC system under study when it is more severe than any other type of contingency. Angular instability is more severe for any S-E1 contingencies that:

- Does not remove other generation near the study generator
- Removes stronger ties from the study generator area to the surrounding BES system.
- Applies the three phase fault to an end of the other Element that is electrically closer to the study generator.

Voltage stability analysis

The S-E1 voltage stability contingencies in the ATC system that are included in study area-centric lists are those for which steady state screening studies indicate the potential for voltage instability.

S-E2. Local or wide area events affecting the Transmission System such as:

S-E2.a (generator 3-phase fault with delayed clearing due to a stuck breaker or a relay failure)

There are no S-E2.a contingencies due to failure of non-redundant relays because all of the BES generators connected to the ATC BES have redundant relays.

Angular stability analysis

No S-E2.a angular stability stuck breaker contingencies in the ATC system are included in the generator-centric lists because any generator under study is more stable when another generator is outaged in the study generator area.

Voltage stability analysis

The S-E2.a voltage stability contingencies in the ATC system that are included in study area-centric lists are those for which steady state screening studies indicate the potential for voltage instability.

S-E2.b (transmission circuit 3-phase fault with delayed clearing due to a stuck breaker or a relay failure)

There are no S-E2.b contingencies due to the failure of a non-redundant relay because all of the ATC BES lines have redundant relays.

Angular stability analysis

The S-E2.b angular stability stuck breaker contingencies included in generator-centric lists are those that will cause more severe angular instability for a BES generator connected to the ATC system under

study when it is more severe than any other type of contingency. Angular instability is more severe for the generator under study for any contingencies that apply a 3-phase fault to the end of any BES line that is electrically close the study generator or remove a stronger tie from the study generator area to the surrounding BES network, but without outaging other generators.

Voltage stability analysis

The S-E2.b voltage stability contingencies in the ATC system that are included in study area-centric lists are those for which steady state screening studies indicate the potential for voltage instability.

S-E2.c (transformer 3-phase fault with delayed clearing due to a stuck breaker or a relay failure)

There are no S-E2.c contingencies due to the failure of a non-redundant relays because all of the ATC BES transformers have redundant relays, except for the Oak Creek 230 kV/138 kV transformer. The Oak Creek transformer S-E2.c relay failure contingency is included in generator stability studies for the Oak Creek 138 kV connected generator, Oak Creek 230 kV connected generators, or Elm Road 345 kV connected generators. Because of the abundance of generation and network interconnectivity in the local area, there is presently no need for relay failure voltage stability studies in this local area.

Angular stability analysis

The S-E2.c angular stability stuck breaker contingencies in the ATC system are included in generator-centric lists are those that will cause more severe angular instability for a BES generator connected to the ATC system under study when it is more severe than any other type of contingency. Angular instability is more severe for the generator under study for any contingencies that apply a 3-phase fault to the end of an ATC BES transformer that is electrically closer the study generator or remove a stronger tie from the area of the study generator to the surrounding BES network, but without outaging other generators.

Voltage stability analysis

The S-E2.c voltage stability contingencies in the ATC system that are included in study area-centric lists are those for which steady state screening studies indicate the potential for voltage instability.

S-E2.d (bus section 3-phase fault with delayed clearing due to a stuck breaker failure or a relay failure)

There are no S-E2.d contingencies due to failure of non-redundant relays for most ATC EHV bus sections, except Arcadian 345 kV, Granville 345 kV, Point Beach 345 kV, and Racine 345 kV buses. In addition, there are no S-E2.d contingencies for ATC HV ring or breaker-and-a-half buses and select HV straight buses.

Angular stability analysis

The S-E2.d angular stability stuck breaker contingencies in the ATC system are included in generator-centric lists are those that will cause more severe angular instability for a BES generator connected to the ATC system under study when it is more severe than any other type of contingency. Angular instability is more severe for the generator under study for any contingencies that will apply a 3-phase fault to an ATC BES bus section that is electrically closer to the study generator or remove stronger ties from the area of the study generator to the surrounding BES network, but without outaging other generators.

Voltage stability analysis

The S-E2.d voltage stability contingencies in the ATC system that are included in study area-centric lists are those for which steady state screening studies indicate the potential for voltage instability.

S-E2.e (3-phase internal breaker fault with normal clearing)

These contingencies represent a 3-phase internal fault on any transmission breaker with normal clearing in the ATC system.

Angular stability analysis

The S-E2.e angular stability contingencies in the ATC system are included in generator-centric lists are those that will cause more severe angular instability for a BES generator connected to the ATC system under study when it is more severe than any other type of contingency. Angular instability is more severe for the generator under study for any contingencies that will apply a 3-phase fault to an ATC BES breaker that is electrically closer to the study generator or remove stronger ties from the study area to the surrounding BES network, but without outaging other generators.

Voltage stability analysis

The S-E2.e voltage stability contingencies in the ATC system that are included in study area-centric lists are those for which steady state screening studies indicate the potential for voltage instability.

S-E2.f Other events based upon operating experience, such as consideration of initiating events that experience suggests may result in wide area disturbances

ATC understands this to mean that these contingencies represent any stability operating experience events that have resulted in the loss of 1000 MW or more of firm load. The 1000 MW threshold is the planning horizon IROL criteria for ATC and MISO.

ATC will consider operating experiences over the previous 10 years.

4. Supporting Information (if applicable)

4.1 Definitions

For NERC defined terms, refer to the latest NERC Glossary of Terms

Bus-Tie Breaker - A circuit breaker that is positioned to connect two individual substation bus configurations (NERC Bus-Tie Breaker Definition). In more explanatory terms, a bus-tie breaker is a breaker positioned between two buses where all sources of fault current into each bus must have their own primary protective device. Capacitor banks, distribution transformers and radial lines to load are not considered to be sources of fault current. In addition, bus-tie breakers are typically breakers that are part of a bus differential scheme. If a bus breaker is part of the line or transformer protection, it is not considered a bus-tie breaker.

Generator - A generator can be a single generator unit or a combination of multiple generator units that are tripped together due to direct or dependent relationships (not including loss of a common cooling system or gas pipeline). These could include, but are not limited to, the following:

- All generating units at a wind farm
- A set of combined cycle generating units where an outage of one generator in the set causes all, or some, of the other generators to be included in outage
- A set of cross-compound generating units where an outage of one generator in the set causes all, or some, of the other generators to be included in outage.

Generating station – Generally, separate generating stations are defined by separate station names that are assigned to them by their Generator Owners. A generating station includes all of the generating units, auxiliary loads, and GSU transformers.

Event-based contingency – Simulate all of the elements outaged by system protection, which could be breaker-to-breaker for lines and transformers or main breaker for generators and shunt devices) contingencies

Individual Substation Bus Configuration - An individual substation bus configuration is a substation electrical bus that is connected to at least two network circuits within the substation and every circuit is connected to the bus through fault interrupting device (e.g. circuit breaker, selected circuit switchers) at the bus. If the terminal of any network circuit is connected to the substation bus without a fault interrupting device, the bus is a circuit terminal and not an individual substation bus configuration.

Power System Circuit - A circuit is a branch for the flow of electrical power (line, transformer, generator, series capacitor, series inductor, or series electronic power flow controller). A network circuit is a branch that has an electrical source of power at both ends of the circuit and therefore network electrical power flow. A radial circuit is a branch that has an electrical source of power at only one end and delivers electrical power to the other end.

Total Load-at-Risk - Total Load-at-Risk is the sum of the following three types of load loss, which are identified and recorded in the steady state multiple outage screening and cascading studies. The definitions are:

- 1) **Consequential Load Loss** (NERC definition)
This is a NERC defined term. The NERC definition is “All Load that is no longer served by the Transmission system as a result of Transmission Facilities being removed from service by a Protection System operation designed to isolate the fault”. Basically, it is the load outaged as a result of the initial event-based outages.
- 2) **Subsequent Cascading Load Loss**
The load outaged as a result of the subsequent element-based outages during the remaining cascading tier analysis. This type of load loss is a result of either load tripping when load buses are below 0.8 p.u. or load islanding due to lines/transformers tripping.
- 3) **Expected Load Shed**
The sum of the load curtailment that may need to be performed by manual or automatic control to restore the ATC system to be within emergency limits and the additional load curtailment that may need to be performed by manual or automatic control to restore the ATC system to be within normal limits.]

4.2 **Reference Documents** - None

4.3 **Workflow** - None

4.4 **Training** - None

4.5 **Related Procedures (ATC or External)** - None

5. Administration (if applicable)

5.1 **Review** - This document will be reviewed within five years of the last revision.

5.2 **Training** - None

5.3 **Testing** - None

5.4 **Retention** - The previous version of this document will be retained for at least five years

after is becomes retired

6. Approvals/Revision History

Revision	Author(s)	Reviewer(s)	Approver(s)	Approval Date	Summary of Changes
0	D. Cullum, R. Morton	- - -	D. Smith	11/18/2010	Initial version
1	D. Cullum, R. Morton	- - -	D. Smith	7/28/2011	Defined severe outage as requiring 100 MW or more load shed
2	D. Cullum, R. Morton, N. Giffin	- - -	D. Smith, P. Walter	9/2012	Changes simulation interval from 5 to 3 years. Added protection failure scenarios. Provided fault examples.
3	C. Lawrence R. Morton, D. Sommer	D. Smith, P. Walter	A. Dolan	12/18/2015	Replaced TPL-002-0, TPL-003-0, and TPL-004-0 documents with version to address TPL-001-4.
4	C. Lawrence R. Morton, D. Sommer	D. Smith, P. Walter	A. Dolan	12/ /2018	Removed the 'load loss allowed' contingencies analysis (i.e. "more severe" contingencies definition and added the use of MISO's MTEP extreme event contingencies analysis within the ATC system

7. Appendix – HVDC Facility Information

The Mackinac back-to-back HVDC facility converts 138 kV AC power to 87 kV DC power and back to 138 kV AC power. This HVDC converter facility is a combination of both series and shunt power control elements. The facility is able to provide 200MW of series real power flow control and two independent 100 MVAR shunt reactive power controls.

The facility does not contain a HVDC transmission line. The back-to-back 87 kV DC converter equipment and associated buses are wholly internal to the HVDC converter and located inside the converter building.

The Table 1 of TPL-001-4 does not explicitly refer to BES FACTS contingency elements (such as SVCs, STATCOMs, back-to-back HVDCs, etc.). So for contingency list development purposes to assess reliable performance of the transmission system, the Mackinac HDVC facility is included in both the TPL-001-4 contingency events that involve transmission lines and ones that involve shunt devices.

The Mackinac HVDC facility contingencies are included in TPL-001-4 contingency events that involve transmission lines because the facility is an element in series with both the MKNG-71 and MKNG-131 138 kV network lines. The outage of the HVDC facility can be represented by three different ways – outage of the entire facility (both converters), outage of the south converter (north converter in shunt

device mode), and outage of the north convertor (south convertor in shunt device mode). Each of these three contingencies simulate the loss of series network line flow.

The Mackinac HVDC facility contingencies are also included in TPL-001-4 contingency events that involve shunt devices because the facility can act as two different shunt reactive resources on either end of the converter.